Grounding LLMs in a Formal Ontology

A pervasive knowledge-graph binding that makes *every* model measurably smarter

F1 0.37 $\rightarrow$ **0.81** across 5 LLMs

VisionFlow ▪ VisionClaw ▪ Agentbox

http://www.visionflow.info

The headline

Connecting a **formal ontology** (4,196 OWL classes, 222k inferred axioms) to *every* AI call lifts factual recall **across the board**.

+0.44 mean F1

Augmented **0.81** vs. ungrounded **0.37** ▪ hallucination roughly **halved**

5 models ▪ 16 KG-grounded questions ▪ 160 isolated runs ▪ objective scoring

Lead, not buried: the binding works, and it is model-agnostic.

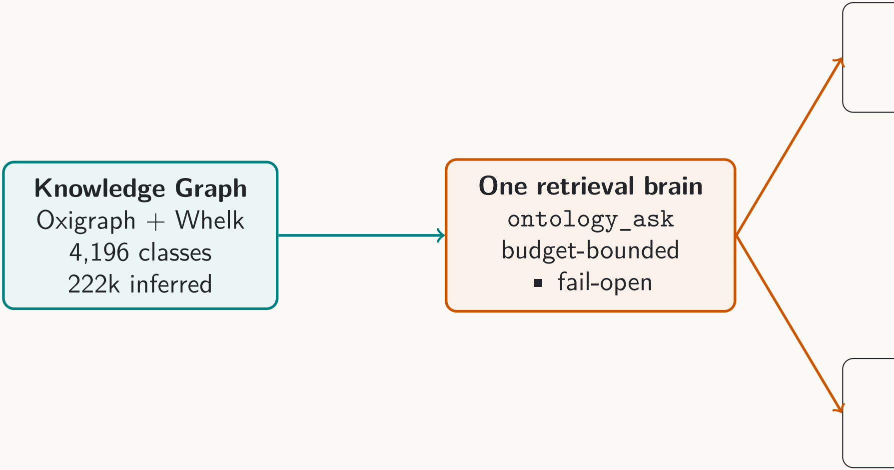
What we built

- **VisionFlow** — the immersive 3D knowledge-graph + agent platform (`visionflow.info`).
- **VisionClaw** — the Rust engine: Oxigraph/Whelk ontology store, GPU physics, real-time graph.
- **Agentbox** — the sovereign agent runtime; 100+ skills, MCP tooling, governed memory.

The final piece: a *pervasive ontology binding* so any AI call can ground itself in the formal knowledge graph — read-pervasive, write-governed, budget-bounded, fail-open.

Features are legion; this deck leads with the one that ties them together.

The binding, in one picture



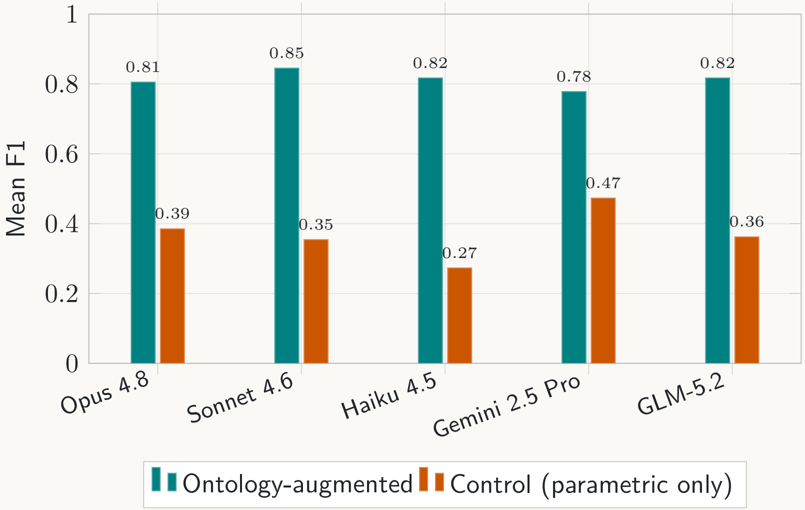
- **Read-pervasive:** every agent, consultant and turn can consult the KG.
- **Write-governed:** proposals are auth-gated and queued; derived facts are fenced.

How we measured it (objectively)

- **KG-as-oracle:** ground truth generated *from the graph itself* — neighbours, subclasses, class existence — so scoring is deterministic, not subjective.
- **Clean A/B:** each cell is an *isolated* session given only the question; augmented arm receives the ontology subgraph, control uses parametric knowledge only.
- **5 models $\times$ 16 questions $\times$ 2 conditions $=$ 160 isolated runs**.
- **Grader:** precision / recall / F1 + hallucination, token-set matched.

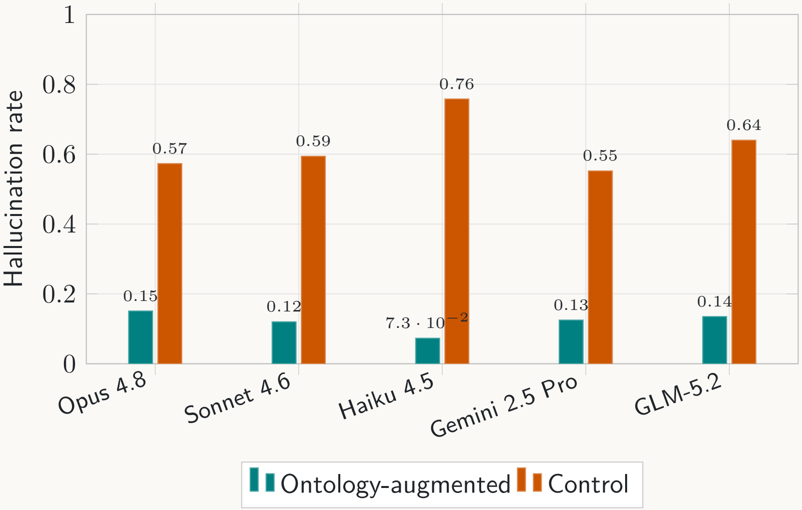
Anthropic Opus/Sonnet/Haiku, Google Gemini 2.5 Pro, Z.AI GLM-5.2.

Result: every model wins



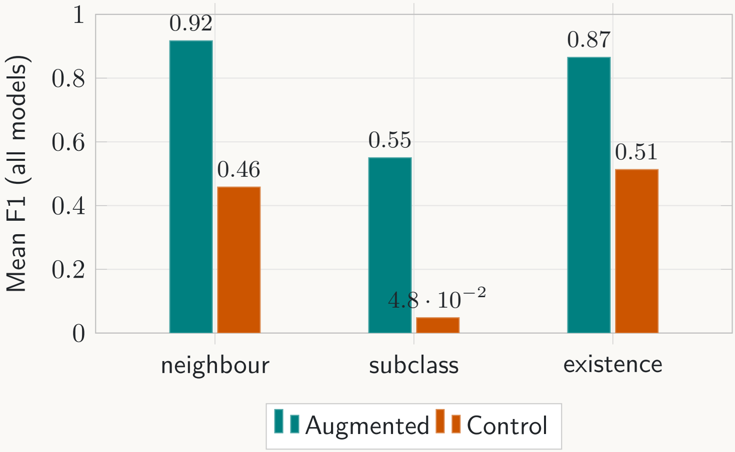
Universal lift: +0.31 to +0.54 F1. The smallest model (Haiku) gains the most.

Result: hallucination roughly halved



Grounding replaces plausible-but-wrong guesses with the graph's actual vocabulary.

Where grounding helps most



Biggest gains on proprietary structure (subclasses: +0.50) and niche concepts the base model can't know.

What it costs

- Grounding adds context tokens and one retrieval round-trip — **optional and per-call**.
- Gemini 2.5 Pro: 4198 prompt tokens/query; GLM-5.2: 3620 — modest for the accuracy gained.
- **Budget-bounded & fail-open:** if the graph is unreachable, the turn proceeds ungrounded — never blocked.

Net: a bounded, switchable cost for a large, universal accuracy gain.

The binding augments thinking without overpowering the context window.

The design works

A formal ontology, bound pervasively to AI, makes **every** model more accurate and less hallucinatory.

- **+0.44 mean F1** across 5 LLMs; hallucination roughly halved.
- Read-pervasive, write-governed, budget-bounded, fail-open — production-shaped.
- One shared brain across tool, consultant and CLI surfaces.

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